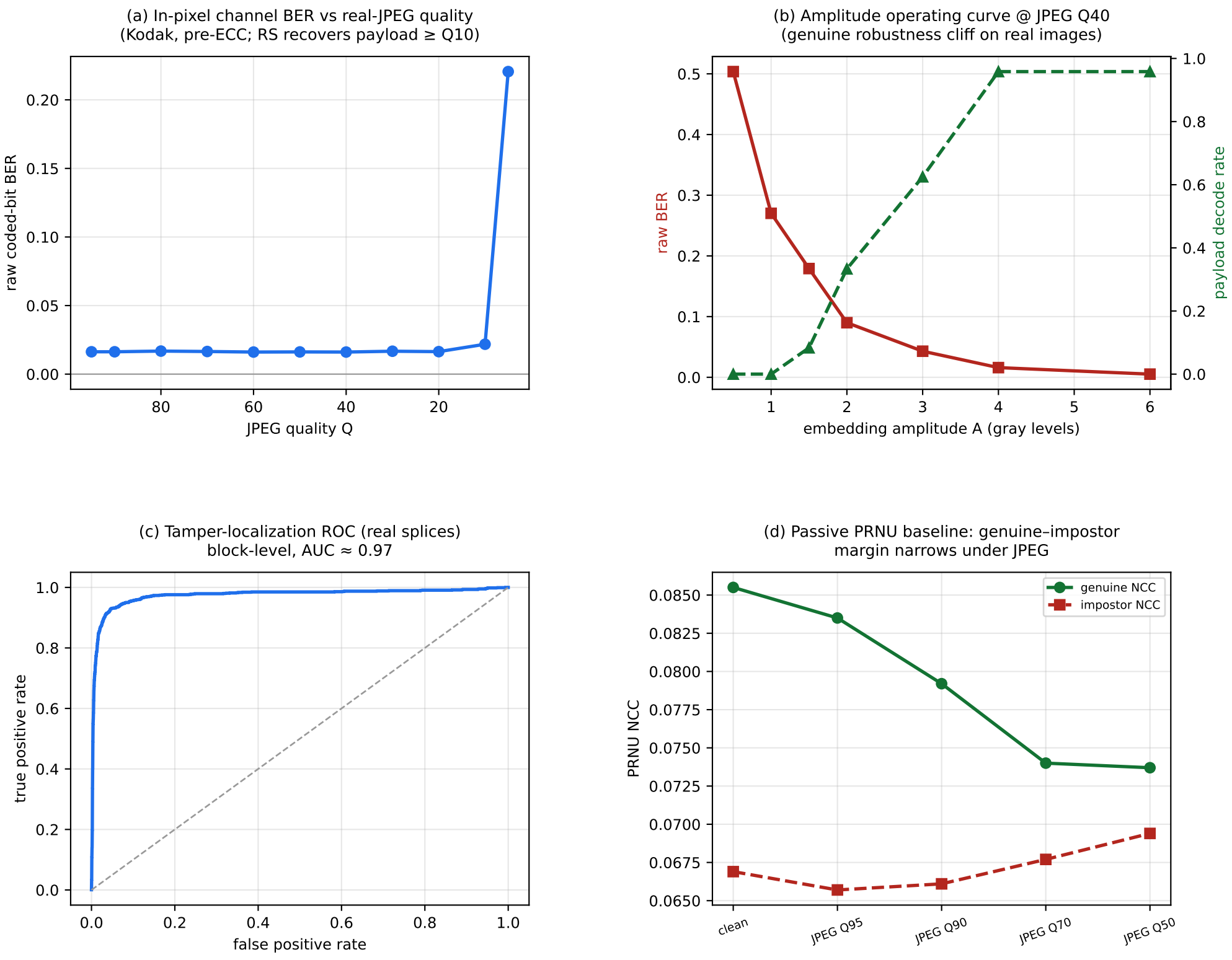
(a) In-pixel channel BER vs real-JPEG quality
(Kodak, pre-ECC; RS recovers payload ≥ Q10)
raw coded-bit BER
0.20
0.15
0.10
0.05
0.00
80
60
40
20
JPEG quality Q
(b) Amplitude operating curve @ JPEG Q40
(genuine robustness cliff on real images)
raw BER
payload decode rate
0.5
0.4
0.3
0.2
0.1
0.0
1.0
0.8
0.6
0.4
0.2
0.0
1
2
3
4
5
6
embedding amplitude A (gray levels)
(c) Tamper-localization ROC (real splices)
block-level, AUC ≈ 0.97
true positive rate
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
false positive rate
(d) Passive PRNU baseline: genuine–impostor
margin narrows under JPEG
PRNU NCC
0.0850
0.0825
0.0800
0.0775
0.0750
0.0725
0.0700
0.0675
0.0650
genuine NCC
impostor NCC
clean
JPEG Q95
JPEG Q90
JPEG Q70
JPEG Q50